

Test 1
grupa 131

2. ~~fil { $f: \mathbb{Z}_m \rightarrow \mathbb{Z}_n \mid f(\bar{x} + \bar{y}) = f(\bar{x}) + f(\bar{y}), f(\bar{0}) = \bar{0} \}$~~

fil { $f: \mathbb{Z}_{12} \rightarrow \mathbb{Z}_{16} \mid f(\bar{x} + \bar{y}) = f(\bar{x}) + f(\bar{y}), f(\bar{0}) = \bar{0} \}$

not $f(\bar{1}) = \bar{a}, \bar{a} \in \mathbb{Z}_{16}$

$$\begin{aligned} f(\bar{x}) &= f(\underbrace{\bar{1} + \dots + \bar{1}}_{x \text{ ori}}) \\ &= \underbrace{f(\bar{1}) + f(\bar{1}) + \dots + f(\bar{1})}_{de x \text{ ori}} \end{aligned}$$

$$= x \cdot f(\bar{1}), x \in \mathbb{Z}_{12}$$

$$= x \cdot \bar{a}, x \in \mathbb{Z}_{12}$$

$$\bar{0} = f(\bar{0}) = f(\bar{12}) = 12 \bar{a} \Rightarrow 16 \mid 12a$$

$$\text{cum } (12, 16) = 4 \Rightarrow 4 \mid a \Rightarrow a = 4k$$

$$\bar{a} = \{ \bar{0}, \bar{4}, \bar{8}, \bar{12} \}$$

$$f_1(\bar{x}) = \bar{0}$$

$$f_2(\bar{x}) = \bar{4}x$$

$$f_3(\bar{x}) = \bar{8}x$$

$$f_4(\bar{x}) = \bar{12}x$$

Shor $K_1 = \mathbb{Z}_{12}$, ~~Im $f_1 = \{ \bar{0} \}$~~

Shor $K_2 = \{ \bar{0}, \bar{4}, \bar{8} \}$

Shor $K_3 = \{ \bar{0}, \bar{2}, \bar{4}, \bar{6}, \bar{8}, \bar{10} \}$

Shor $K_4 = \{ \bar{0}, \bar{4}, \bar{8} \}$

$$\text{Im } f_1 = \{ \bar{0} \}$$

$$\text{Im } f_2 = \{ \bar{0}, \bar{4}, \bar{8}, \bar{12} \}$$

$$\text{Im } f_3 = \{ \bar{0}, \bar{8} \}$$

$$\text{Im } f_4 = \{ \bar{0}, \bar{4}, \bar{8}, \bar{12} \}$$

$$1. (i) U(\mathbb{Z}_{15}) = \{x \in \mathbb{Z}_{15} \mid x \text{ invertibil}\}$$

$$U(\mathbb{Z}_{15}) = \{x \in \mathbb{Z}_{15} \mid (x, 15) = 1\}$$

$$U(\mathbb{Z}_{15}) = \{\hat{1}, \hat{2}, \hat{4}, \hat{7}, \hat{8}, \hat{11}, \hat{13}, \hat{14}\}$$

$$\text{ord}(U(\mathbb{Z}_{15})) = 8$$

$$(ii) \text{ord}(U(\mathbb{Z}_{15})) = 8 \Rightarrow (\exists) \hat{x} \in \mathbb{Z}_{15} \text{ a c. } U(\mathbb{Z}_{15}) = \langle \hat{x} \rangle$$

$$\text{ord}(\hat{x}) = 8 \Rightarrow U(\mathbb{Z}_{15}) = \langle \hat{x} \rangle$$

$$U(\mathbb{Z}_{15}) = (\hat{1}, \hat{x}, \hat{x}^2, \hat{x}^3, \hat{x}^4, \hat{x}^5, \hat{x}^6)$$

$$\Rightarrow U(\mathbb{Z}_{15}) \text{ ciclic}$$

$$\text{iar } \mathbb{Z}_2 \times \mathbb{Z}_4 \text{ este ciclic generat } (\bar{1}, \bar{1}) >, \bar{1} \in \mathbb{Z}_2, \bar{1} \in \mathbb{Z}_4$$

$$\Rightarrow \text{cum ambele grupuri sunt ciclice cu 8 elem} \Rightarrow$$

$$U(\mathbb{Z}_{15}) \cong \mathbb{Z}_2 \times \mathbb{Z}_4$$